

Homework 2 - MA 513 - Due Sun 2026-2-08

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1

1.1 a)

$$z^4 + 1 = 0$$

$$z^4 = -1$$

$$z^4 = e^{i\pi(2n+1)} \quad n \in \mathbb{Z}$$

$$z = e^{i\pi \frac{(2n+1)}{4}}$$

$$z = e^{i\frac{\pi}{4}}, e^{i\frac{3\pi}{4}}, e^{i\frac{5\pi}{4}}, e^{i\frac{7\pi}{4}}$$

$$z = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i, -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i, -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i, \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$$

1.2 b)

$$z^6 = 1 + i$$

$$z^6 = \frac{\sqrt{2}}{2} \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right)$$

$$z^6 = \frac{\sqrt{2}}{2} e^{i\pi(\frac{1}{4}+2n)} \quad n \in \mathbb{Z}$$

$$z = \frac{\sqrt{2}}{2} e^{i\pi(\frac{1+8n}{24})}$$

$$z = \sqrt[6]{\frac{\sqrt{2}}{2}} e^{i\frac{\pi}{24}}, \sqrt[6]{\frac{\sqrt{2}}{2}} e^{i\frac{9\pi}{24}}, \sqrt[6]{\frac{\sqrt{2}}{2}} e^{i\frac{17\pi}{24}}, \sqrt[6]{\frac{\sqrt{2}}{2}} e^{i\frac{25\pi}{24}}, \sqrt[6]{\frac{\sqrt{2}}{2}} e^{i\frac{33\pi}{24}}, \sqrt[6]{\frac{\sqrt{2}}{2}} e^{i\frac{41\pi}{24}}$$

$$z \approx 0.9358 + 0.1232i, 0.3612 + 0.872i, -0.5746 + 0.7488i, -0.9358 - 0.1232i, -0.3612 - 0.872i, 0.5746 - 0.7488i$$

2

2.1 a)

$$\begin{aligned}
 z &= x + yi \\
 \operatorname{Re}\left(\frac{i}{z}\right) &= 2 \\
 \operatorname{Re}\left(\frac{i \cdot \bar{z}}{z \cdot \bar{z}}\right) &= \\
 \operatorname{Re}\left(\frac{ix + y}{|z|^2}\right) &= \\
 \operatorname{Re}\left(\frac{y}{x^2 + y^2} + \frac{ix}{x^2 + y^2}\right) &= \\
 \frac{y}{x^2 + y^2} &= 2 \\
 y &= 2x^2 + 2y^2 \\
 0 &= x^2 + y^2 - \frac{1}{2}y \\
 \frac{1}{16} &= x^2 + \left(y - \frac{1}{4}\right)^2
 \end{aligned}$$

This gives a circle with radius $\frac{1}{4}$ and a center at $(0, \frac{1}{4})$

2.2 b)

$$\begin{aligned}
 \operatorname{Re}(z^2) &= 1 \\
 \operatorname{Re}((x + yi)^2) &= 1 \\
 \operatorname{Re}(x^2 - y^2 + 2xyi) &= 1 \\
 x^2 - y^2 &= 1
 \end{aligned}$$

This gives a hyperbola with asymptotes at $y = \pm x$ and a center at $(0, 0)$

3

3.1 a)

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{n^2 + i}{1 - in^2} &= \lim_{n \rightarrow \infty} \frac{(n^2 + i) \cdot (1 + in^2)}{(1 - in^2) \cdot (1 + in^2)} \\
 &= \lim_{n \rightarrow \infty} \frac{n^2 + i + in^4 - n^2}{1 + n^4} \\
 &= \lim_{n \rightarrow \infty} \frac{i(1 + n^4)}{1 + n^4} \\
 &= \lim_{n \rightarrow \infty} i \\
 &= i
 \end{aligned}$$

3.2 b)

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{n^2 - 1}{n(n + 2)} e^{i\pi \frac{n}{n+1}} &= \lim_{n \rightarrow \infty} \frac{n^2 - 1}{n^2 + 2n} \cdot \lim_{n \rightarrow \infty} (e^{i\pi})^{\frac{n}{n+1}} \\
 &= 1 \cdot \lim_{n \rightarrow \infty} (-1)^{\frac{n}{n+1}} \\
 &= \lim_{n \rightarrow \infty} (-1)^{\frac{1}{1+\frac{1}{n}}} \\
 &= (-1)^{\frac{1}{1}} \\
 &= -1
 \end{aligned}$$

3.3 c)

$$\begin{aligned}
 a &= re^{i\pi\theta} \\
 \lim_{n \rightarrow \infty} a^n &= \lim_{n \rightarrow \infty} r^n e^{in\pi\theta}
 \end{aligned}$$

If $|r| < 1$ then $\lim_{n \rightarrow \infty} a^n = \lim_{n \rightarrow \infty} r^n e^{in\pi\theta} = 0$. If $|r| = 1$ then $\lim_{n \rightarrow \infty} a^n = \lim_{n \rightarrow \infty} r^n e^{in\pi\theta} = \text{undefined}$. If $|r| > 1$ then $\lim_{n \rightarrow \infty} a^n = \lim_{n \rightarrow \infty} r^n e^{in\pi\theta} = \infty$. So if $|a| < 1$ then

$$\lim_{n \rightarrow \infty} \frac{a^n}{1 + a^{2n}} = a^n \cdot \frac{1}{1 + a^{2n}} = 0 \cdot 1.$$

If $|a| = 0$ then

$$\begin{aligned}
\lim_{n \rightarrow \infty} \frac{a^n}{1 + a^{2n}} &= \lim_{n \rightarrow \infty} \frac{e^{2\pi i n}}{1 + e^{4\pi i n}} \\
&= \lim_{n \rightarrow \infty} \frac{1}{e^{-2\pi i n} + e^{2\pi i n}} \\
&= \lim_{n \rightarrow \infty} \frac{1}{\cos(-2\pi n) + \sin(-2\pi n) + \cos(2\pi n) + \sin(2\pi n)} \\
&= \lim_{n \rightarrow \infty} \frac{1}{\cos(2\pi n) + -\sin(2\pi n) + \cos(2\pi n) + \sin(2\pi n)} \\
&= \lim_{n \rightarrow \infty} \frac{1}{2 \cos(2\pi n)} \\
&= \lim_{n \rightarrow \infty} \frac{1}{2} \sec(2\pi n) = \text{undetermined}
\end{aligned}$$

If $|a| > 0$ then

$$\begin{aligned}
\lim_{n \rightarrow \infty} \frac{a^n}{1 + a^{2n}} &= \lim_{n \rightarrow \infty} \frac{a^{-n}}{a^{-2n} + 1} \\
&= \lim_{n \rightarrow \infty} \frac{1}{a^n} \cdot \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{a^{2n}} + 1} \\
&= 0 \cdot \frac{1}{1} = 0
\end{aligned}$$

4

4.1 a)

$$\begin{aligned}
\lim_{z \rightarrow 0} \frac{|z|^2}{z} &= \lim_{z \rightarrow 0} \frac{z \bar{z}}{z} \\
&= \lim_{z \rightarrow 0} \bar{z} \\
&= \lim_{z \rightarrow 0} (x - yi)
\end{aligned}$$

Assuming $0 < \delta < \epsilon$, the limit is 0, and $\epsilon > |z - 0|$ then

$$\begin{aligned}
 \delta &> |f(z) - 0| = |(x - yi) - 0| \\
 &= |x - yi| \\
 &= (x^2 + y^2)^2 \\
 &> |x + yi| \\
 &= |z - 0| \\
 |z - 0| &< \delta < \epsilon.
 \end{aligned}$$

Which means that

$$\lim_{z \rightarrow 0} \frac{|z|^2}{z} = 0$$

4.2 b)

$$\begin{aligned}
 \lim_{z \rightarrow 0} \frac{\bar{z}}{z} &= \lim_{z \rightarrow 0} \frac{x - yi}{x + yi} \\
 \lim_{x \rightarrow 0} \left(\frac{x - yi}{x + yi} \right) &= \frac{-yi}{y} = -1 \\
 \lim_{y \rightarrow 0} \left(\frac{x - yi}{x + yi} \right) &= \frac{x}{x} = 1 \\
 -1 &\neq 1
 \end{aligned}$$

Therefore the limit does not exist.

4.3 c)

$$\begin{aligned}
 \lim_{z \rightarrow 0} \frac{z + 1}{|z| + 1} &= \lim_{z \rightarrow 0} \frac{x + yi + 1}{\sqrt{x^2 + y^2} + 1} \\
 \lim_{y \rightarrow 0} \lim_{x \rightarrow 0} \frac{x + yi + 1}{\sqrt{x^2 + y^2} + 1} &= \lim_{y \rightarrow 0} \frac{yi + 1}{y + 1} = \frac{1}{1} \\
 \lim_{x \rightarrow 0} \lim_{y \rightarrow 0} \frac{x + yi + 1}{\sqrt{x^2 + y^2} + 1} &= \lim_{x \rightarrow 0} \frac{x + 1}{x + 1} = \frac{1}{1} \\
 \text{let } x &= y \\
 \lim_{x \rightarrow 0} \frac{x + xi + 1}{\sqrt{x^2 + x^2} + 1} &= \frac{1}{-1} = -1 \\
 -1 &\neq 1
 \end{aligned}$$

Therefore the limit does not exist.

4.4 d)

$$\lim_{z \rightarrow 0} \frac{\operatorname{Re}(z)\operatorname{Im}(z)}{|z|} = \lim_{z \rightarrow 0} \frac{xy}{\sqrt{x^2 + y^2}}$$

$$\lim_{x \rightarrow 0} \frac{xy}{\sqrt{x^2 + y^2}} = \frac{0}{y} = 0$$

$$\lim_{y \rightarrow 0} \frac{xy}{\sqrt{x^2 + y^2}} = \frac{0}{x} = 0$$

let $x = y$

$$\lim_{x \rightarrow 0} \frac{x^2}{\sqrt{x^2 + x^2}} = \frac{0}{0} \text{ form}$$

$$= \lim_{x \rightarrow 0} \frac{2x\sqrt{2x^2} - \frac{2x^3}{\sqrt{2x^2}}}{2x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{2x^2}}{x} - \frac{x}{\sqrt{2x^2}}$$

$$= \lim_{x \rightarrow 0} \frac{2\sqrt{2x^2}}{2x} - \frac{\sqrt{2x^2}}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{2x^2}}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{4x^2}{\sqrt{2x^2}} - 2\sqrt{2x^2}}{4x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{4x^2}{\sqrt{2x^2}} - 2\sqrt{2x^2}}{4x^2}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\sqrt{2x^2}} - \frac{\sqrt{2x^2}}{2x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{2x^2}}{2x^2} - \frac{\sqrt{2x^2}}{2x^2} = 0$$

$$= \frac{0}{0} \text{ form}$$

$$\lim_{z \rightarrow 0} \frac{\operatorname{Re}(z)\operatorname{Im}(z)}{|z|} = 0$$

5

5.1 a)

Let $\delta < \epsilon$ then

$$\begin{aligned}
 \delta &> |f(z) - f(z_0)| \\
 &= |\bar{z} - z_0| \\
 &\geq ||\bar{z}| - |-z_0|| \\
 &= ||z| - |-z_0|| \\
 &= ||z| + |-z_0| - 2|z_0|| \\
 &\geq ||z - z_0| + |-z_0| - |-z_0|| \\
 &\geq ||z - z_0| - |0|| \\
 |z - z_0| &\leq \delta < \epsilon
 \end{aligned}$$

Therefore $f(z)$ is continuous.

5.2 b)

Let $\delta < \epsilon$ then

$$\begin{aligned}
 \delta &> |f(z) - f(z_0)| \\
 &= |Im(z) - z_0| \\
 &= |y - z_0| \\
 &= |-i(x + yi) + ix - z_0| \\
 &= |-i(z) + ix - z_0| \\
 &\geq ||-iz + ix| - |-z_0|| \\
 &\geq |||z| - |x|| - |-z_0|| \\
 &= |||z| + |x| - |x| - |-x|| + |-z_0| - 2|-z_0|| \\
 &\geq |||z + x| + |-z_0| - |-z_0| - |z_0|| \\
 &\geq ||z + x - z_0|| \\
 &\geq |z - z_0 + x| \\
 &\geq ||z - z_0| - |x|| \\
 &= ||z - z_0| + |-x| - |-x| - |x|| \\
 &\geq |z - z_0 - x|
 \end{aligned}$$

$$\begin{aligned}
 |z - z_0 - x| &\leq |z - z_0| \leq |z - z_0 + x| < \delta < \epsilon \\
 |z - z_0| &\leq \delta < \epsilon
 \end{aligned}$$

Therefore $f(z)$ is continuous.

5.3 c)

Let $\delta < \epsilon$ then

$$\begin{aligned}
\delta &> |f(z) - f(z_0)| \\
&= |Re(z^2) - z_0| \\
&= |x^2 - y^2 - z_0| \\
&\geq ||x^2 - y^2| - |-z_0|| \\
&= ||x^2 + y^2 - 2y^2| - |-z_0|| \\
&= ||(z)(x - yi) - 2y^2| + |-z_0| - 2|z_0|| \\
&\geq ||(z)(x - yi)| - |-2y^2|| + |-z_0| - 2|z_0|| \\
&= ||(z)\bar{z}| - |-y^2| - |y^2|| + |-z_0| - 2|z_0|| \\
&\geq |||z||z| + |-z_0|| \\
&\geq |z^2 - z_0| \\
&= |x^2 + 2xy + y^2 - z_0| \\
&\geq ||x^2 + y^2 - z_0| - |2xy|| \\
&= |||x^2 + y^2| - |-z_0|| - |-xy| + |xy|| \\
&\geq ||z| - |-z_0|| \\
&= ||z| + |-z_0| - 2|-z_0|| \\
&\geq ||z - z_0| - |-z_0| - |z_0|| \\
&\geq |z - z_0|
\end{aligned}$$

$$|z - z_0| \leq \delta < \epsilon$$

Therefore $f(z)$ is continuous.

6

$$\begin{aligned}
 c &= e^{\pi i \frac{2k}{n}} \quad k \in \{1 \dots n-1\} \\
 1 + c + c^2 + \dots + c^{n-1} &= \frac{1 - z^n}{1 - z} \\
 &= \frac{1 - \left(e^{\pi i \frac{2k}{n}}\right)^n}{1 - e^{\pi i \frac{2k}{n}}} \\
 &= \frac{1 - e^{\pi i \frac{2k}{n}(n)}}{1 - e^{\pi i \frac{2k}{n}}} \\
 &= \frac{1 - e^{2k\pi i}}{1 - e^{\pi i \frac{2k}{n}}} \\
 &= \frac{1 - 1}{1 - e^{\pi i \frac{2k}{n}}} \\
 &= 0
 \end{aligned}$$

7

7.1 a)

$$\begin{aligned}
 h &= h_x + h_y i \\
 \lim_{h \rightarrow 0} \frac{Re(z+h) - Re(z)}{h} &= \lim_{h \rightarrow 0} \frac{x + h_x - x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{h_x}{h_x + h_y i} \\
 \lim_{h_x \rightarrow 0} \frac{h_x}{h_x + h_y i} &= \frac{0}{h_y i} = 0 \\
 \lim_{h_y \rightarrow 0} \frac{h_x}{h_x + h_y i} &= \frac{h_x}{h_x} = 1 \\
 0 &\neq -1
 \end{aligned}$$

Therefore the limit does not exist and $Re(z)$ is not differentiable.

7.2 b)

$$\begin{aligned}
 h &= h_x + h_y i \\
 \lim_{h \rightarrow 0} \frac{\operatorname{Im}(z+h) - \operatorname{Im}(z)}{h} &= \lim_{h \rightarrow 0} \frac{y + h_y - y}{h} \\
 &= \lim_{h \rightarrow 0} \frac{h_y}{h_x + h_y i} \\
 \lim_{h_x \rightarrow 0} \frac{h_y}{h_x + h_y i} &= \frac{0}{h_x} = 0 \\
 \lim_{h_y \rightarrow 0} \frac{h_y}{h_x + h_y i} &= \frac{h_y}{h_y i} = -i \\
 0 &\neq -i
 \end{aligned}$$

Therefore the limit does not exist and $\operatorname{Im}(z)$ is not differentiable.

8

8.1 a)

$f(z)$ is analytic everywhere since $f(z) = z^8 + 7z^5 - \pi z^2 + 1 \in P(x)$.

8.2 b)

$f(z)$ is analytic everywhere except $z = 0, 1, 2$ since $f(z)$ blows to infinity at these points.

8.3 c)

$f(z)$ is analytic everywhere except $z = e^{\frac{2k\pi}{5}i}, k \in \{0, 1, 2, 3, 4\} = 1, 0.309 + 0.9511i, -0.809 + 0.5878i, -0.809 - 0.5878i, 0.309 - 0.9511i$ since $f(z)$ blows to infinity at these points.

9

9.1 i)

$$f(z) = \frac{1}{z^2 + 1} \quad (1)$$

This function is not analytical at $z = i, -i$ as the equation blows to infinity but is analytical everywhere else.

9.2 ii)

$$f(z) = z(z-1)(z-2)(z-3)(z-4)(z-5) = z^6 - 15z^5 + 85z^4 - 225z^3 + 274z^2 - 120z \quad (2)$$

This function is analytical at all points in $\mathbb{C}$ but its zeros make its reciprocal not analytical at $z = 0, 1, 2, 3, 4, 5$